PCOSence

PCOS Risk Prediction System

"Empowering Women Through Early PCOS Detection with Artificial Intelligence."

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ML MICROPROJECT

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The Problem: PCOS Goes Undetected

Many women don't know they have PCOS because its symptoms are often ignored or misunderstood. Delayed diagnosis can lead to serious health problems.

🌍 Widespread

Affects millions of women globally across all age groups and demographics

⚠ Overlooked

Varying symptoms cause many women to ignore early warning signs

🔬 Hard to Diagnose

Requires specialist access and extensive testing often delayed for years

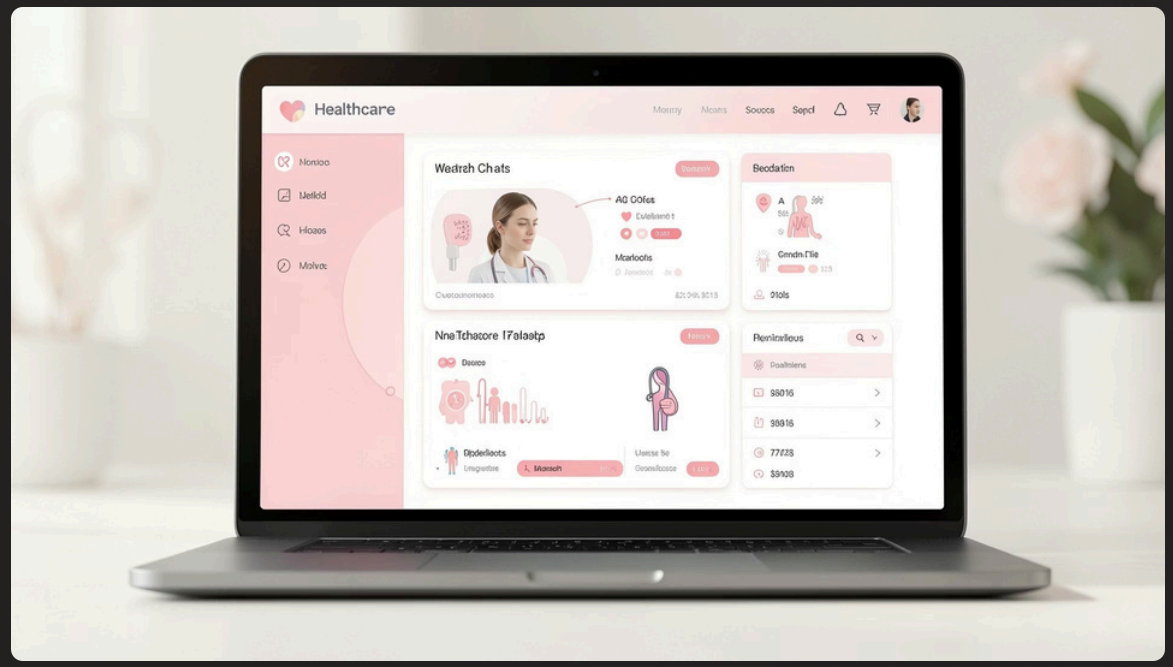
🩺 Serious Consequences

Infertility, diabetes, obesity, hormonal imbalance, and mental health issues



Introducing PCOSence

PCOSence uses Artificial Intelligence and Machine Learning to predict the risk of PCOS and provide a fast, user-friendly preliminary health assessment for women.



Early Risk Prediction

ML-powered assessment from user health data

Personalized Recommendations

Diet, exercise, and doctor guidance tailored to results

User-FriendlyInterface

Intuitive design accessible on any device

Future Hospital Integration

Seamless connection to healthcare providers

Project Objectives

PCOSence is built on six main goals that combine technology with women's healthcare.

Predict with ML

Accurate PCOS risk prediction using trained machine learning m o dels

Raise Awareness

Educate women about PCOS symptoms and early warning signs

FastScreening

Provide quick preliminary assessment without specialist visits

Reduce Delay

Minimize late diagnosis through proactive early detection

Promote Wellness

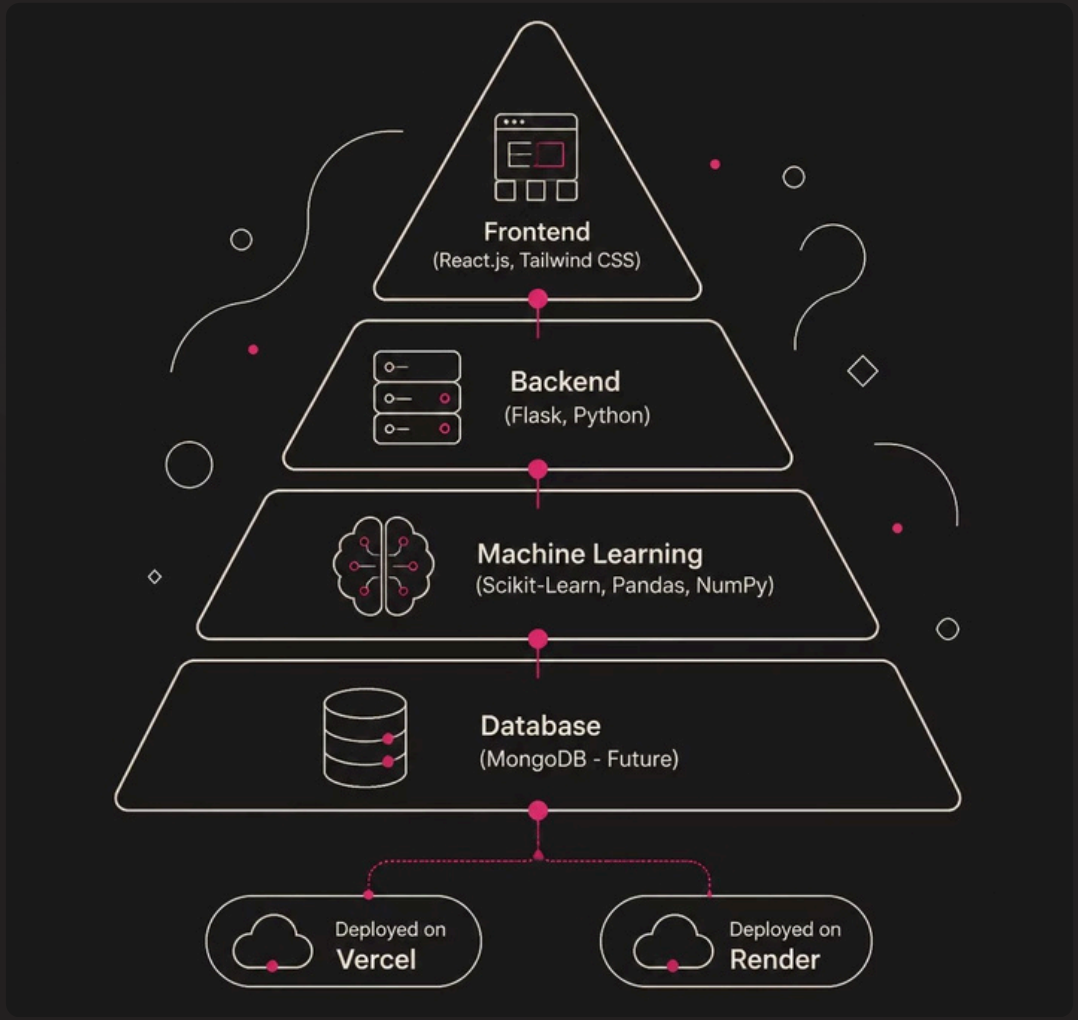
Deliver personalized diet and exercise recommendations

Accessible Platform

Build an inclusive, mobile-friendly healthcare tool for all

Technology Stack

PCOSence is built using modern technologies, with a responsive frontend and an intelligent Machine Learning backend.



Frontend
React.js , CSS

Backend

Flask Python

Machine Learning

Scikit-Learn Pandas NumPy

Database (Future)

Mon goDB

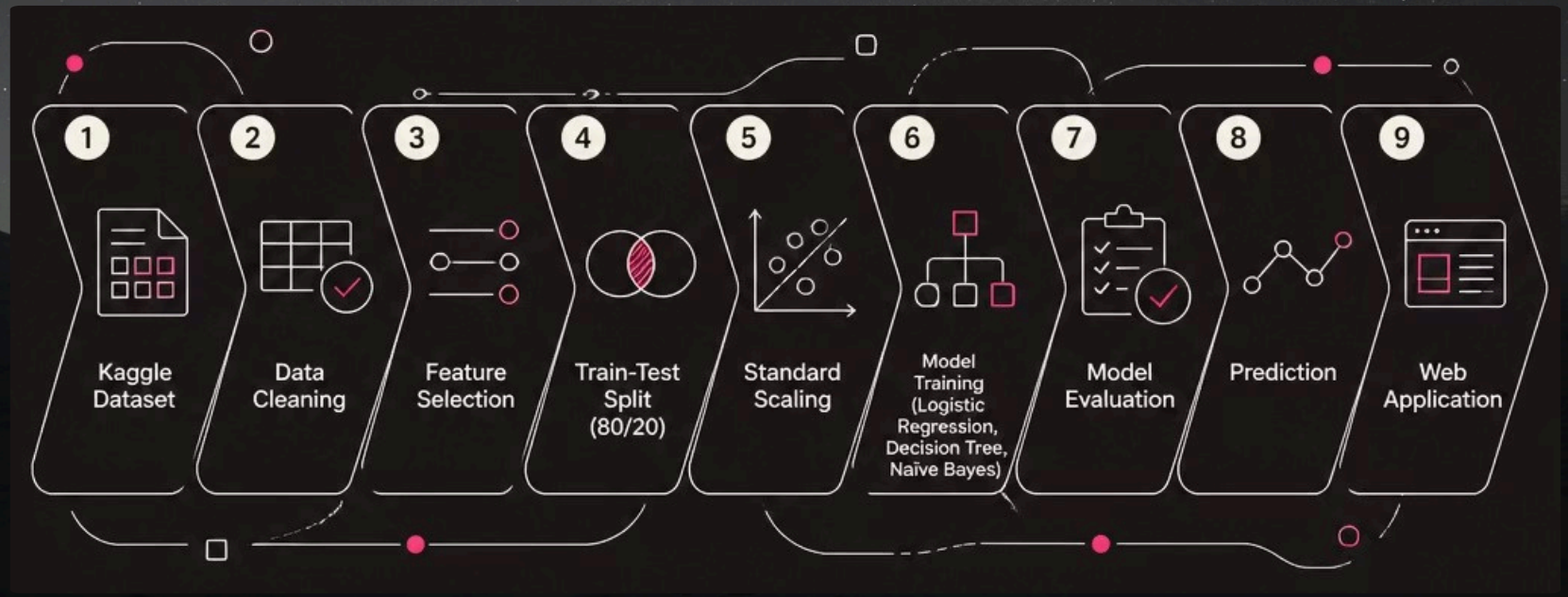
Deployment

Vercel Render

Dataset & ML Workflow

Kaggle PCOS Dataset

The system uses a PCOS dataset containing important health information such as age, BMI, weight, height, menstrual cycle details, acne, hair growth, hair loss, weight gain, exercise habits, and lifestyle factors.



Target Variable: PCOS — Yes / No

Data Preprocessing & ML Models

Preprocessing Pipeline

01

Remove Duplicates

Eliminate redundant records

02

Handle Missing Values

Impute or remove incomplete data

03

Feature Selection

Identify clinically relevant attributes

04

Normalization & Split

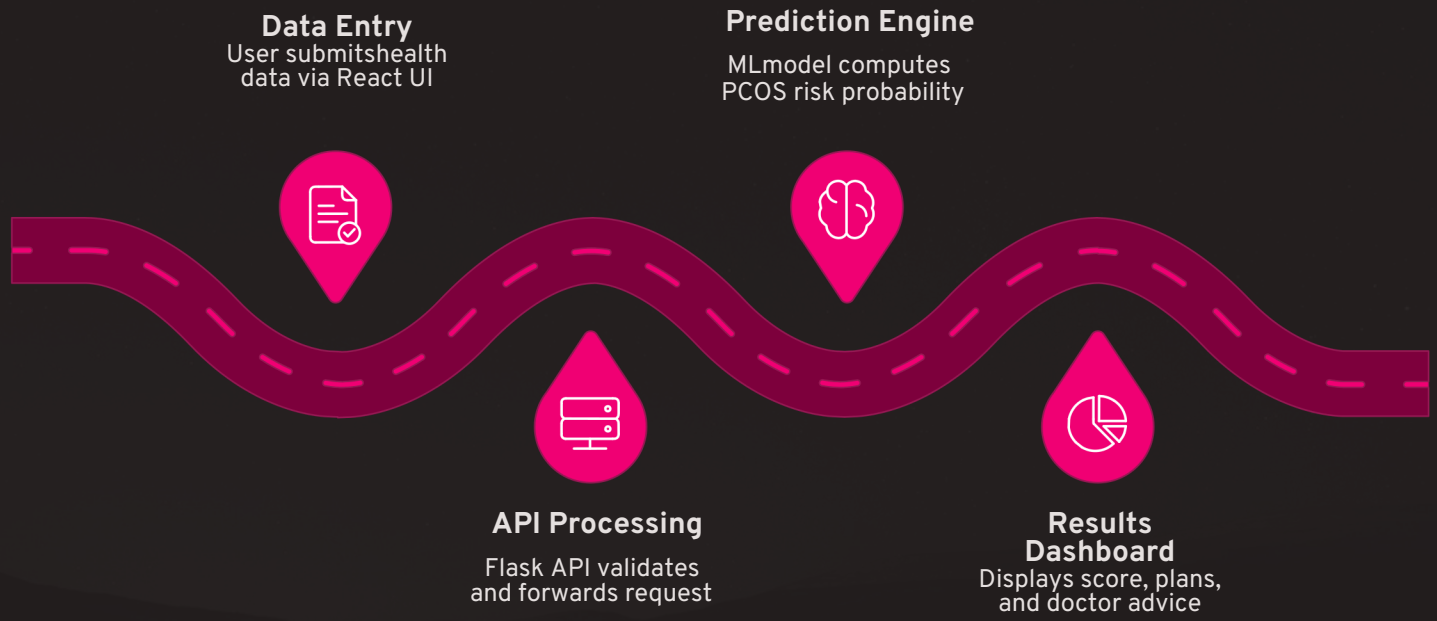
Scale data, 80/20 train-test split

Model Comparison

Model	Speed	Accuracy	Interpretability
Logistic Regression ✓	Fast	High	Excellent
Decision Tree	Moderate	High	Good
Naïve Bayes	Very Fast	Moderate	Moderate

Logistic Regression selected as the primary model for its balance of accuracy, speed, and clinical interpretability.

System Architecture & User Workflow



The platform delivers a seamless experience — from data input to personalized health guidance — in seconds.

🔮 PCOS Prediction

AI risk assessment with probability score

📊 Health Report

Downloadable report with BMI calculator

🥗 Diet & Exercise

Personalized lifestyle planners

📱 Responsive Design

Mobile-friendly, accessible anywhere



Future Scope & Advantages

1 — Phase 1

Hospital integration,online appointment booking ,nearby hospital locator

2 — Phase 2

AIhealth chatbot, period tracker, menstrual cycle prediction

3 — Phase 3

Wearable integration, medicine reminders, cloud database, multi-language support

Early Detection

Catch PCOS before complications arise

Affordable & Fast

Screening without costly specialist visits

Personalized

Tailored recommendations for every user

Accessible

Available anywhere,on any device

Conclusion

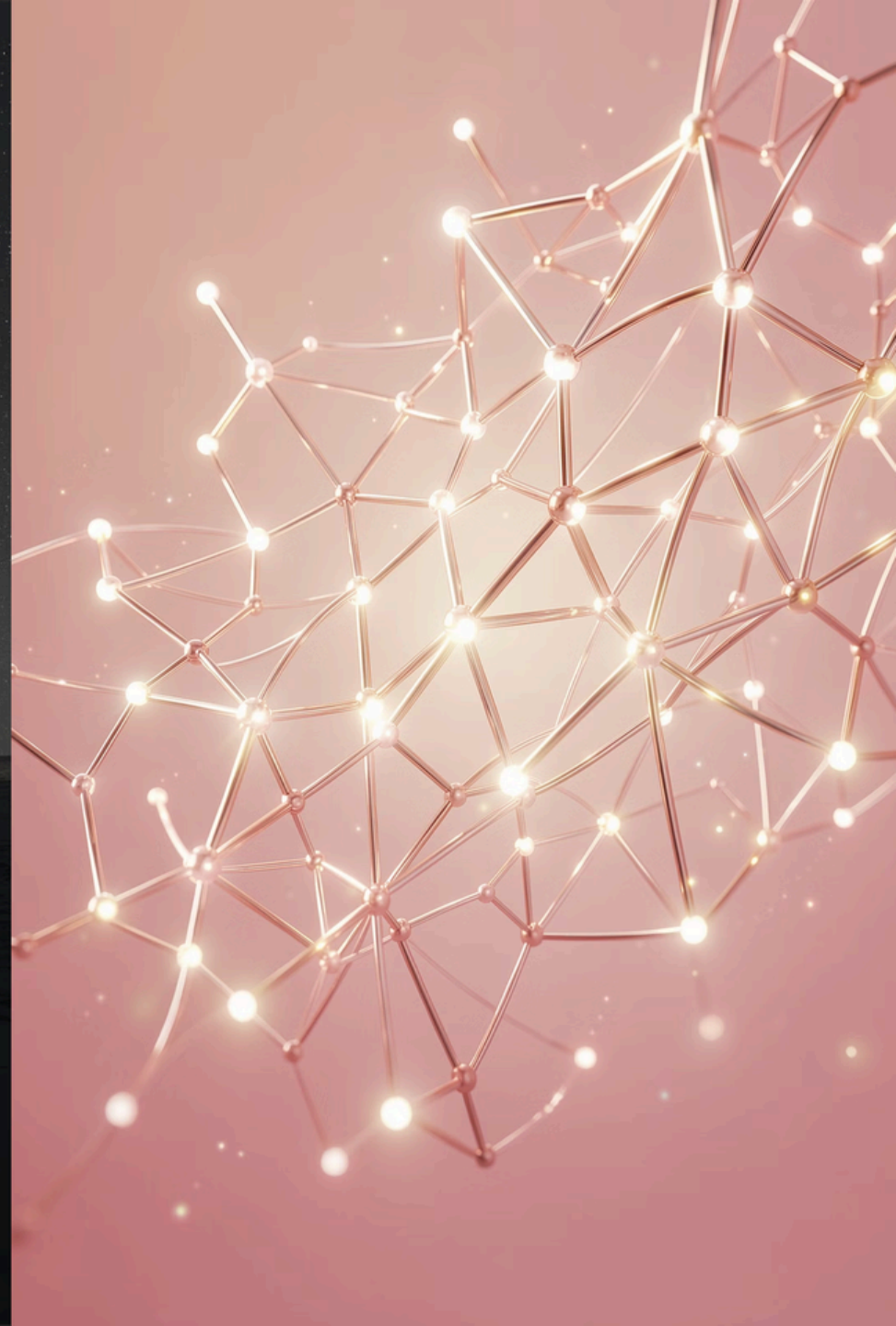
PCOSence uses Artificial Intelligence and Machine Learning to predict the risk of PCOS at an early stage. It helps women by providing health insights, diet and exercise recommendations, and encourages timely medical consultation.

🤝 Thank You

Questions & Answers Welcome

🎓 Project

Third Year Engineering



THANK YOU EVERYONE